

# Observability assurance report

Module 10 | Monitoring, distributed tracing and incident correlation

## EXECUTIVE SUMMARY

The implementation meets the Module 10 acceptance objective. RED metrics, guarded alerts, distributed traces and structured logs form a coherent investigation path from a service symptom to its controlled root cause.

Retained deployment evidence confirms the alert lifecycle, cross-tool identifier matching and return to a healthy state. The design also bounds metric cardinality and excludes sensitive request data from telemetry.

## KEY FINDINGS

- 01

Actionable alerting

Grafana evaluates guarded error-rate and latency rules every 10 seconds; a breach must persist for 10 minutes.
- 02

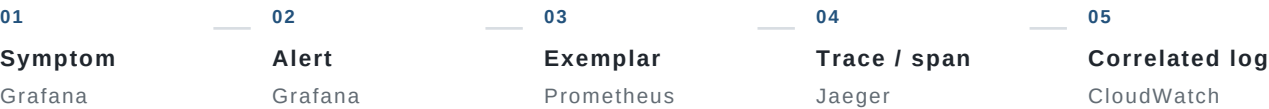
End-to-end correlation

The exemplar trace ID resolves to Jaeger; the trace and span IDs then locate the exact CloudWatch JSON log.
- 03

Bounded telemetry

Metric labels and log fields are allowlisted, while trace export is credential-free and limited to a private receiver.

## INVESTIGATION PATH



Correlation key    Sampled W3C trace ID; span ID identifies the exact operation.

## RED SIGNAL AND ALERT POLICY

| SIGNAL   | MEASUREMENT                       | POLICY                             |
|----------|-----------------------------------|------------------------------------|
| Rate     | Total requests                    | Dashboard baseline                 |
| Errors   | 5xx / all requests over 5 minutes | > 5%; 20+ requests; 10 minutes     |
| Duration | p95 histogram over 5 minutes      | > 300 ms; 20+ requests; 10 minutes |

**CONTROL NOTE**        Labels are limited to method, route and status. Logs exclude bodies, headers, cookies, tokens, query values and arbitrary errors.

# Evidence and conclusion

Controlled validation, correlation chain and acceptance boundary

## CONTROLLED VALIDATION

| TEST                                              | OBSERVED EFFECT                                         | VERIFICATION AND RECOVERY                                                                      |
|---------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Error mode</b><br>POST /test   value=error     | <b>Deliberate HTTP 500</b><br>Assessed as a bounded run | Pending, Firing and Resolved states captured.<br>Stopping controlled traffic restored service. |
| <b>Latency mode</b><br>POST /test   value=latency | <b>Fixed 400 ms success</b><br>Assessed independently   | Alert behavior verified without suppression, bypass or configuration change.                   |

## CORRELATION CHAIN

|                                  |                             |                                |                                |                                 |
|----------------------------------|-----------------------------|--------------------------------|--------------------------------|---------------------------------|
| 01                               | 02                          | 03                             | 04                             | 05                              |
| <b>Alert state</b><br>UTC marker | <b>Exemplar</b><br>Trace ID | <b>Jaeger</b><br>Matching span | <b>CloudWatch</b><br>Exact IDs | <b>Root cause</b><br>/test mode |

The same identifier links the metric exemplar, Jaeger trace and structured log. Route-stable span names and the selected test mode establish causality.

## VERIFICATION SEQUENCE

|    |                     |                                                           |
|----|---------------------|-----------------------------------------------------------|
| 01 | <b>Baseline</b>     | Healthy scrape state and 10-minute normal baseline.       |
| 02 | <b>Error mode</b>   | Bounded 500 traffic; full alert lifecycle captured.       |
| 03 | <b>Latency mode</b> | Fixed 400 ms successes assessed separately.               |
| 04 | <b>Correlate</b>    | Exemplar, spans and exact log IDs matched.                |
| 05 | <b>Recover</b>      | Ordinary traffic, versions and healthy state reconfirmed. |

## ACCEPTANCE BOUNDARY

### PROVEN IN REPOSITORY

HTTP server/client spans, RED metrics and exemplars; Jaeger and Grafana provisioning; both alert rules.

### VERIFIED IN DEPLOYMENT

Populated panels, alert lifecycles, trace links and exact log IDs; recovery, ordinary traces, health and versions.

|                   |                                                                                     |
|-------------------|-------------------------------------------------------------------------------------|
| <b>CONCLUSION</b> | <b>Acceptance evidence is complete and the service returned to a healthy state.</b> |
| <b>LIMITS</b>     | Non-durable lab Jaeger; 100% lab sampling; database tracing not applicable.         |